

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250270066

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

IIDA; Yosuke

AUTOMATIC DOCUMENT FEEDER AND IMAGE FORMING APPARATUS

Abstract

An automatic document feeder includes a document stacker, a document conveyor, a sound generator, a sound collector, and circuitry. The document stacker is a receptacle on which a document is placed. The document conveyor conveys the document on the document stacker along a document conveyance path in a conveyance direction. The sound generator is on a surface of the document conveyance path, and has a protrusion or a recess protruding or recessed from the surface of the document conveyance path, and extending continuously or intermittently from one end to another end of the document conveyance path in a width direction orthogonal to the conveyance direction. The sound collector collects sound generated by contact between the sound generator and a fastening object attached to the document. The circuitry is to predict whether the fastening object is attached to the document, based on the sound collected by the sound collector.

Inventors: IIDA; Yosuke (Kanagawa, JP)

Applicant: IIDA; Yosuke (Kanagawa, JP)

Family ID: 1000008487086

Assignee: Ricoh Company, Ltd. (Tokyo, JP)

Appl. No.: 19/053889

Filed: February 14, 2025

Foreign Application Priority Data

JP 2024-029075

Feb. 28, 2024

Publication Classification

Int. Cl.: B65H29/22 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2024-029075, filed on Feb. 28, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] Embodiments of the present disclosure relate to an automatic document feeder and an image forming apparatus.

Background Art

[0003] Typical automatic document feeders continuously feeds multiple sheet-like documents placed on a document tray one by one toward an image reading position of a scanner as a document reader. Such automatic document feeders have been considered to determine a jam (paper jam) or a wrinkle by sound. For example, a medium feeding device in the art is disclosed to include a sound detector disposed downstream from a feeder in a feeding direction and determine occurrence of a jam when the sound volume exceeds a threshold value. In addition, the medium feeding device is also disclosed to include a contactor and detect wrinkles by detecting sound generated by contact of a medium and the contactor.

[0004] However, the medium feeding device in the art does not have a countermeasure to prevent a possible cause of jam when a fastening object such as a clip is attached to the original document. Due to such a configuration, when an original document is conveyed with a fastening object attached to the original document, no sound occurs after the fastening object passes by the contactor, and the medium feeding device in the art fails to detect the fastening object.

SUMMARY

[0005] Embodiments of the present disclosure described herein provide a novel automatic document feeder including a document stacker, a document conveyor, a sound generator, a sound collector, and circuitry. The document stacker is a receptacle on which a document is placed. The document conveyor conveys the document on the document stacker along a document conveyance path in a conveyance direction. The sound generator is disposed on a surface of the document conveyance path. The sound generator has a protrusion or a recess protruding or recessed from the surface of the document conveyance path, and extending continuously or intermittently from one end to another end of the document conveyance path in a width direction orthogonal to the conveyance direction. The sound collector collects sound generated by contact between the sound generator and a fastening object attached to the document. The circuitry is to predict whether the fastening object is attached to the document, based on the sound collected by the sound collector.

[0006] Further, embodiments of the present disclosure described herein provide an image forming apparatus including the above-described automatic document feeder, and an image forming device to form an image on a medium.

[0007] Further, embodiments of the present disclosure described herein provide an automatic document feeder including a document path, a protrusion, and a sound collector. The document conveyance path is a path in which a document is conveyed to a reading position to read the document. The protrusion extends in a direction orthogonal to a conveyance direction of the document in the document conveyance path and protrudes upward toward a downstream side from

an upstream side in the document conveyance path of the conveyance direction of the document, to contact a surface of the document that conveys in the document conveyance path. The sound collector collects sound of the document that conveys in the document conveyance path. Conveyance of the document to be conveyed to the reading position is stopped based on the sound collected by the sound collector.

[0008] Further, embodiments of the present disclosure described herein provide an image forming apparatus including the above-described automatic document feeder, and an image forming device to form an image on a medium.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] Exemplary embodiments of this disclosure will be described in detail based on the following figures, wherein:

[0010] FIG. 1 is a schematic diagram illustrating a configuration of an image forming apparatus according to an embodiment of the present disclosure;

[0011] FIG. 2 is a partially enlarged view of a part of an image forming device according to an embodiment of the present disclosure;

[0012] FIG. 3 is a partially enlarged view of a part of a tandem section including four image forming units of an image forming device according to an embodiment of the present disclosure;

[0013] FIG. 4 is a perspective view of a scanner and an automatic document feeder according to an embodiment of the present disclosure;

[0014] FIG. 5 is a perspective view of an automatic document feeder according to an embodiment of the present disclosure;

[0015] FIG. 6 is a schematic diagram illustrating a configuration of an automatic document feeder according to an embodiment of the present disclosure;

[0016] FIG. 7 is a partially enlarged view of a main part of the automatic document feeder of FIG. 5;

[0017] FIG. 8 is a top view of a part of a document conveyance path according to an embodiment of the present disclosure;

[0018] FIG. 9 is a perspective view of a sound generator according to an embodiment of the present disclosure;

[0019] FIG. 10 is a perspective view of a sound generator according to a first modification of an embodiment of the present disclosure;

[0020] FIG. 11 is a schematic diagram illustrating an example of an original document to which a fastening object is attached;

[0021] FIG. 12 is a diagram illustrating conveyance of the original document of FIG. 11;

[0022] FIG. 13 is a schematic diagram illustrating another example of an original document to which a fastening object is attached;

[0023] FIG. 14 is a diagram illustrating conveyance of the original document of FIG. 13;

[0024] FIG. 15 is a perspective view of a sound generator according to a second modification of an embodiment of the present disclosure;

[0025] FIG. 16 is a perspective view of a sound generator according to a third modification of an embodiment of the present disclosure;

[0026] FIG. 17 is a perspective view of a sound generator according to a fourth modification of an embodiment of the present disclosure;

[0027] FIG. 18 is a top view of a part of the document conveyance path according to the first modification of an embodiment of the present disclosure;

[0028] FIG. 19 is a diagram illustrating the part of the document conveyance path of FIG. 18,

viewed from the upstream side in the conveyance direction;

[0029] FIG. **20** is a side view of the part of the document conveyance path of FIG. **18**;

[0030] FIG. **21** is a perspective view of a sound generator according to a fifth modification of an embodiment of the present disclosure;

[0031] FIG. **22** is a top view of a part of the document conveyance path according to the second modification of an embodiment of the present disclosure;

[0032] FIG. **23** is an enlarged perspective view of a part of the document conveyance path of [0033] FIG. **22**;

[0034] FIG. **24** is a top view of a part of the document conveyance path of the automatic document feeder according to another example of an embodiment of the present disclosure;

[0035] FIG. **25** is a block diagram illustrating a part of an electric circuit of an image forming apparatus according to an embodiment of the present disclosure; and

[0036] FIG. **26** is a flowchart of a fastening object determination process executed by a controller according to an embodiment of the present disclosure.

[0037] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted.

DETAILED DESCRIPTION

[0038] It will be understood that if an element or layer is referred to as being “on,” “against,” “connected to” or “coupled to” another element or layer, then it can be directly on, against, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, if an element is referred to as being “directly on,” “directly connected to” or “directly coupled to” another element or layer, then there are no intervening elements or layers present. As used herein, the term “connected/coupled” includes both direct connections and connections in which there are one or more intermediate connecting elements. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0039] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, term such as “below” can encompass both an orientation of above and below.] The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors herein interpreted accordingly.

[0040] The terminology used herein is for describing particular embodiments and examples and is not intended to be limiting of exemplary embodiments of this disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0041] Embodiments of the present disclosure are described below in detail with reference to the drawings. Like reference signs are assigned to identical or equivalent components and a description of those components may be simplified or omitted.

[0042] The following describes an image forming apparatus according to an embodiment of the present disclosure. Descriptions are given of an image forming apparatus, for example, an electrophotographic copier, in the following embodiments, according to an embodiment of the present disclosure.

[0043] A basic structure of the image forming apparatus according to the present embodiment is firstly described.

[0044] FIG. 1 is a schematic diagram illustrating a configuration of an image forming apparatus **100** according to an embodiment of the present disclosure.

[0045] As illustrated in FIG. 1, the image forming apparatus **100** includes an image forming device **1** that forms an image on a sheet-like medium, a sheet feeding device **40**, and an image reading system **50**. The image reading system **50** includes a scanner **150** as an image reading device that is fixed on the image forming device **1**, and an automatic document feeder **51** (ADF **51**) supported by the scanner **150**.

[0046] The sheet feeding device **40** includes a paper bank **41**, multiple sheet trays **42** disposed one above the other in the paper bank **41**, sheet feed rollers **43** each of which picking up a recording sheet from a selected one of the multiple sheet trays **42**, and a sheet separation roller **45** that separates the recording sheets fed by the sheet feed rollers **43**. The sheet feeding device **40** further includes multiple conveyance roller pairs **46** that convey the recording sheet in a sheet conveyance path **37** as a conveyance path in the image forming device **1**.

[0047] The multiple sheet trays **42** contains multiple recording sheets overlapping each other in a form of a sheet bundle. The sheet feed roller **43** presses and contacts the uppermost sheet of the multiple recording sheets on each of the multiple sheet trays **42**. As the sheet feed roller **43** rotates, the uppermost recording sheet of the sheet bundle is fed from the selected one of the sheet trays **42**.

[0048] The multiple conveyance roller pairs **46** are disposed near the multiple sheet trays **42**. Each of the multiple conveyance roller pairs **46** includes a first conveyance roller and a second conveyance roller adjacent to (on the right side of FIG. 1) the first conveyance roller. The first conveyance roller and the second conveyance roller of each of the multiple conveyance roller pairs **46** are in contact with each other to form a conveyance nip region.

[0049] A sheet separation roller **45** is disposed below the first conveyance roller of each of the multiple conveyance roller pairs **46** and is in contact with the first conveyance roller from below to form a separation conveyance nip region.

[0050] A recording sheet fed from one of the sheet trays **42** driven and rotated by a corresponding one of the sheet feed rollers **43** enters the separation conveyance nip region formed by the contact of the first conveyance roller of each of the multiple conveyance roller pairs **46** and the sheet separation roller **45** disposed below the first conveyance roller. In the separation conveyance nip region, the first conveyance roller that contacts the upper face of the recording sheet applies a conveyance force to the recording sheet from the sheet tray **42** toward a sheet feeding path **44** as the first conveyance roller is driven and rotated in the counterclockwise direction in FIG. 1. In contrast, the sheet separation roller **45** that is in contact with the lower face of the recording sheet applies a conveyance force to the recording sheet from the sheet feeding path **44** toward the sheet tray **42** as the sheet separation roller **45** is driven and rotated in the counterclockwise direction in FIG. 1, thereby returning the recording sheet to the sheet tray **42**.

[0051] When only one recording sheet is fed from the sheet tray **42**, the first conveyance roller and the sheet separation roller **45** apply the conveyance force to the recording sheet toward opposite directions to each other in the separation conveyance nip region. As a result, a load exceeding a given threshold value is applied to the drive transmission part of the sheet separation roller **45**. Then, a torque limiter disposed in the drive transmission part of the sheet separation roller **45** is operated to cut off the transmission of the driving force from a DC brushless motor to the sheet separation roller **45**. Accordingly, the sheet separation roller **45** is rotated with the recording sheet that is conveyed by the first conveyance roller, and the recording sheet is then ejected from the separation conveyance nip region to the sheet feeding path **44**.

[0052] On the other hand, when the multiple recording sheets are fed from the sheet tray **42** with the multiple recording sheets overlapped to each other, the first conveyance roller applies the conveyance force to the uppermost recording sheet of the multiple recording sheets from the sheet

tray **42** toward the sheet feeding path **44** in the separation conveyance nip region. Then, the uppermost recording sheet of the multiple recording sheets is fed from the separation conveyance nip region toward the sheet feeding path **44**. On the other hand, the sheet separation roller **45** applies the conveyance force from the sheet feeding path **44** toward the sheet tray **42** to the lower recording sheet or sheets of the multiple recording sheets, so that the lower recording sheet is (or sheets are) reversed from the separation conveyance nip region toward the sheet tray **42**.

Accordingly, in the separation conveyance nip region, the uppermost recording sheet is separated from other recording sheet or sheets so as to be conveyed alone to the sheet feeding path **44**.

[0053] The recording sheet on the sheet feeding path **44** enters the conveyance nip region of the conveyance roller pair **46** where the conveyance force is applied upward from below in the vertical direction. Accordingly, the recording sheet is conveyed toward the sheet conveyance path **37** of the image forming device **1** in the sheet feeding path **44**.

[0054] The image forming device **1** includes an optical writing device **2** and four image forming units **3K**, **3Y**, **3M**, and **3C**. The image forming units **3K**, **3Y**, **3M**, and **3C** form black, yellow, magenta, and cyan toner images, respectively. The image forming device **1** further includes a transfer unit **24**, a sheet conveyance unit **28**, a registration roller pair **33**, a fixing device **34**, a switchback device **36**, and the sheet conveyance path **37**. The optical writing device **2** includes a light source such as a laser diode and a light emitting diode (LED). The light source is disposed in the optical writing device **2**. By driving the light source in the optical writing device **2**, laser lights **L** are emitted toward four drum-shaped photoconductors **4K**, **4Y**, **4M**, and **4C** to irradiate respective surfaces of the drum-shaped photoconductors **4K**, **4Y**, **4M**, and **4C**. Due to this irradiation, electrostatic latent images of respective single colors are formed on the surfaces of the photoconductors **4K**, **4Y**, **4M**, and **4C**, which will be developed to visible toner images via a given development process.

[0055] FIG. **2** is a partially enlarged view of a part of the image forming device **1** according to an embodiment of the present disclosure.

[0056] FIG. **3** is a partially enlarged view of a tandem section including the four image forming units **3K**, **3Y**, **3M**, and **3C** of the image forming device **1** according to an embodiment of the present disclosure.

[0057] Since the four image forming units **3K**, **3Y**, **3M**, and **3C** have respective configurations substantially the same as each other except the toner colors, the image forming units **3K**, **3Y**, **3M**, and **3C** are also described without suffixes indicating the toner colors, which are **K**, **Y**, **M**, and **C** in FIG. **3**. For example, the image forming units **3K**, **3Y**, **3M**, and **3C** are referred to as a “process unit **3**” in a single form occasionally.

[0058] The image forming units **3K**, **3Y**, **3M**, and **3C** also include respective image forming components disposed around each of the photoconductors **4K**, **4Y**, **4M**, and **4C** as a single unit supported by a common support member, respectively. The image forming units **3K**, **3Y**, **3M**, and **3C** are detachably attached to the image forming device **1** of the image forming apparatus **100**. The image forming unit (i.e., the image forming units **3K**, **3Y**, **3M**, and **3C**) includes the photoconductor **4** (i.e., the photoconductors **4K**, **4Y**, **4M**, and **4C**), and a charging device **23**, the developing device **6** (i.e., the developing devices **6K**, **6Y**, **6M**, and **6C**), the drum cleaning device **15** (i.e., the drum cleaning devices **15K**, **15Y**, **15M**, and **15C**), and an electric discharging lamp **22** around the photoconductor **4**. The image forming apparatus **100** is a tandem image forming apparatus in which the four image forming units **3K**, **3Y**, **3M**, and **3C** are aligned in a direction of movement of an intermediate transfer belt **25** as an endless loop.

[0059] The photoconductor **4** is manufactured by a hollow tube made of aluminum, for example, with a drum shape covered by an organic photoconductive layer having photosensitivity. The photoconductor **4** may have a shape of endless belt.

[0060] The developing device **6** develops an electrostatic latent image into a visible toner image by a two-component developer including magnetic carrier particles and non-magnetic toner. The two-

component developer is now referred to as a “developer”. The developing device **6** includes an agitating portion **7** and a development portion **11**. The agitating portion **7** stirs the two-component developer accommodated therein and conveys the two-component developer to a development sleeve **12**. The development portion **11** supplies the non-magnetic toner, which is included in the two-component developer and held by the development sleeve **12**, to the photoconductor **4**.

[0061] The agitating portion **7** is located at a position lower than the development portion **11** and includes two screw, a partition, a development case **9**, and a toner concentration sensor **10**. The two transfer screws **8** are disposed in parallel to each other. The partition is disposed between the two transfer screws **8**. The development case **9** has an opening or a slot to face the photoconductor **4**. The toner concentration sensor **10** is disposed on the bottom of the development case **9**.

[0062] The development portion **11** includes the development sleeve **12**, a magnetic roller **13**, and a doctor blade **14**. The development sleeve **12** faces the photoconductor **4** through the opening (or the slot) of the development case **9**. The magnetic roller **13** is fixedly or unrotatably disposed inside the development sleeve **12**. The doctor blade **14** is disposed adjacent to the development sleeve **12** and the leading end of the doctor blade **14** is disposed close to the development sleeve **12**. The development sleeve **12** has a non-magnetic, rotatable tubular body. The magnetic roller **13** has multiple magnetic poles arranged in the order in a rotation direction of the development sleeve **12**, starting from an opposed position to the doctor blade **14**. Each of these magnetic poles applies a magnetic force at a predetermined position in the rotation direction of the development sleeve **12**, with respect to the two-component developer supplied on the development sleeve **12**. With this action of the magnetic roller **13**, the two-component developer that is conveyed from the agitating portion **7** is attracted and attached to the surface of the development sleeve **12** and a magnetic brush of toner is formed along the lines of the magnetic force on the surface of the development sleeve **12**.

[0063] In accordance with rotation of the development sleeve **12**, the magnetic brush is regulated to have an appropriate layer thickness when passing by the opposed position to the doctor blade **14**. Then, the magnetic brush is moved to a development region facing the photoconductor **4**. Due to a difference of potentials between a development bias that is applied to the development sleeve **12** and an electrostatic latent image formed on the surface of the photoconductor **4**, the toner is transferred onto the electrostatic latent image, so that the electrostatic latent image is developed into a visible toner image. Further, after returning into the development portion **11** again along with the rotation of the development sleeve **12** then leaving from the surface of the development sleeve **12** due to repulsion of the magnetic field formed between the magnetic poles of the magnetic roller **13**, the two-component developer in a form of the magnetic brush is returned to the agitating portion **7**. An appropriate amount of toner is supplied to the two-component developer in the agitating portion **7** based on a result or results detected by the toner concentration sensor **10**. Alternative to the two-component developer, the developing device **6** according to the present embodiment may employ one-component developer that does not include magnetic carriers.

[0064] The drum cleaning device **15** includes a cleaning blade **16**, a fur brush **17**, an electric field roller **18**, a scraper **19**, a collection screw **20**, and an external toner recycling transfer device **21**. The cleaning blade **16** is an elastic member to be pressed against the photoconductor **4**, so as to scrape residual toner remaining on the surface of the photoconductor **4**. In the present embodiment, the drum cleaning device **15** employs a blade member such as the cleaning blade **16**, however, the configuration is not limited thereto. Alternative to the blade member, a brush roller, for example, can be applied to the drum cleaning device **15**. The fur brush **17** according to the present embodiment is provided in order to increase the cleanability. The fur brush **17** is a conductive member and the outer circumferential face of the fur brush **17** slidably contacts the photoconductor **4**. The fur brush **17** according to the present embodiment is rotatable in a direction indicated by arrow in FIG. **3**.

[0065] The fur brush **17** also functions as an applier that scrapes a solid lubricant to obtain fine

powder of lubricant and applies the scraped fine powder to the surface of the photoconductor **4**. The electric field roller **18** is a metallic member that applies a bias to the fur brush **17**. The electric field roller **18** is disposed rotatably in a direction indicated by arrow in FIG. **3**. The scraper **19** has a leading end that is pressed against the electric field roller **18**. The toner removed from the photoconductor **4** and attached to the fur brush **17** is transferred onto the electric field roller **18** that contacts the fur brush **17** in a counter direction to be applied with a bias while the electric field roller **18** is rotating. After being scraped and removed from the electric field roller **18** by the scraper **19**, the toner falls onto the collection screw **20**. The collection screw **20** conveys the toner collected from the surface of the photoconductor **4** toward an end portion of the drum cleaning device **15** in a direction orthogonal to the drawing sheet, and transfers the collected toner to the external toner recycling transfer device **21**. The external toner recycling transfer device **21** sends the collected toner to the developing device **6** for recycling.

[0066] The electric discharging lamp **22** removes residual electric charge remaining on the surface of the photoconductor **4** by photo irradiation. After such residual electric charge is removed, the electrically discharged surface of the photoconductor **4** is uniformly charged by the charging device **23** again and then optically irradiated by the optical writing device **2**. In the image forming apparatus **100** according to the present embodiment, the charging device **23** is a charging roller that is applied with charging bias and rotates while contacting the photoconductor **4**. However, in some embodiments, the charging device **23** may be a scorotron charger that performs a charging process on the photoconductor **4** in non-contact with the photoconductor **4**.

[0067] According to the above-described operations with the configuration illustrated in FIG. **2**, black (K), yellow (Y), magenta (M), and cyan (C) toner images are formed on the photoconductors **4K**, **4Y**, **4M**, and **4C** of the image forming units **3K**, **3Y**, **3M**, and **3C**, respectively. The transfer unit **24** is disposed below the four image forming units **3K**, **3Y**, **3M**, and **3C**.

[0068] The transfer unit **24** endlessly moves the intermediate transfer belt **25** in the clockwise direction in FIG. **2** while the intermediate transfer belt **25** is stretched by and would around multiple rollers and is in contact with the photoconductors **4K**, **4Y**, **4M**, and **4C**. By so doing, respective primary transfer nip regions for forming black, yellow, magenta, and cyan images are formed between the photoconductors **4K**, **4Y**, **4M**, and **4C** and the intermediate transfer belt **25**. In proximity to each of the primary transfer nip regions for black, yellow, magenta, and cyan images, the primary transfer rollers **26** (i.e., the primary transfer rollers **26K**, **26Y**, **26M**, and **26C**) are disposed in contact with the inner loop of the intermediate transfer belt **25** to press the intermediate transfer belt **25** against the photoconductors **4** (i.e., the photoconductors **4K**, **4Y**, **4M**, and **4C**), respectively. A primary transfer bias is applied by respective transfer bias power supplies to the primary transfer rollers **26K**, **26Y**, **26M**, and **26C**. Consequently, respective primary transfer electric fields are generated in the primary transfer nip regions to electrostatically transfer respective black, yellow, magenta, and cyan toner images formed on the photoconductors **4K**, **4Y**, **4M**, and **4C** onto the intermediate transfer belt **25**.

[0069] As the intermediate transfer belt **25** passes through the primary transfer nip regions along the endless rotation in the clockwise direction in FIG. **2**, the black, yellow, magenta, and cyan toner images are sequentially transferred at the primary transfer nip regions and overlaid onto an outer circumferential surface of the intermediate transfer belt **25**. Due to the primary transfer of the toner images, a four-color composite toner image (referred to as a four-color toner image) is formed on the surface of the intermediate transfer belt **25**.

[0070] The sheet conveyance unit **28** is disposed below the transfer unit **24** in FIG. **2**. The sheet conveyance unit **28** includes a sheet transfer belt **29**, a sheet transfer belt drive roller **30**, and a secondary transfer roller **31**. The sheet transfer belt **29** is an endless belt that is wound around the sheet transfer belt drive roller **30** and the secondary transfer roller **31** and rotates in a direction indicated by arrow in FIG. **2**. As illustrated in FIG. **2**, the intermediate transfer belt **25** and the sheet transfer belt **29** are sandwiched between the secondary transfer roller **31** and a lower tension roller

27 of the transfer unit **24**. According to this configuration, a secondary transfer nip region is formed between the surface of the intermediate transfer belt **25** and the surface of the sheet transfer belt **29** contacting with each other. A secondary transfer bias is applied by a transfer bias power source to the secondary transfer roller **31**. On the other hand, the lower tension roller **27** of the transfer unit **24** is electrically grounded. By so doing, a secondary transfer electric field is formed in the secondary transfer nip region.

[0071] The registration roller pair **33** is disposed on a right side of the secondary transfer nip region in FIGS. **2** and **3**. A registration roller sensor is disposed adjacent to an entrance of the registration nip region of the registration roller pair **33**. The recording sheet is conveyed from a sheet feeding device to the registration roller pair **33**. After a given time has elapsed from the detection of the leading end of the recording sheet by the registration roller sensor, the conveyance of the recording sheet temporarily stops, and the leading end of the recording sheet contacts the registration nip region of the registration roller pair **33**.

[0072] After the leading end of the recording sheet contacts the registration nip region of the registration roller pair **33**, the registration roller pair **33** restarts the rotation to synchronize the movement of the recording sheet with the movement of the four-color toner image formed on the intermediate transfer belt **25**. Consequently, the recording sheet nipped between the registration roller pair **33** is conveyed to the secondary transfer nip region. When the four-color toner image formed on the intermediate transfer belt **25** closely contacts the recording sheet at the secondary transfer nip region, the four-color toner image on the intermediate transfer belt **25** is transferred onto the recording sheet in the secondary transfer due to the secondary transfer electric field and the nip pressure. At this time, the four-color toner image is combined with white color of the recording medium to make a full-color image. After passing through the secondary transfer nip region, the recording sheet having the full-color toner image on the surface is separated from the intermediate transfer belt **25**. Then, while being held on the front face of the sheet transfer belt **29**, the recording sheet is conveyed to the fixing device **34** along with endless rotation of the sheet transfer belt **29** in the direction as illustrated in FIG. **3**.

[0073] Residual toner that has not been transferred onto the recording sheet in the secondary transfer nip region remains on the surface of the intermediate transfer belt **25** after the intermediate transfer belt **25** has passed through the secondary transfer nip region. The residual toner is scraped and removed from the surface of the intermediate transfer belt **25** by a belt cleaning device **32** that is disposed in contact with the surface of the intermediate transfer belt **25**.

[0074] The recording sheet is conveyed to the fixing device **34**. The fixing device **34** fixes the full-color toner image to the recording sheet by application of heat and pressure. Then, the recording sheet is conveyed from the fixing device **34** to the sheet ejection roller pair **35** to be ejected to the outside of the image forming apparatus **100**.

[0075] As illustrated in FIG. **1**, the switchback device **36** is disposed below the sheet conveyance unit **28** and the fixing device **34**. As a result of the above-described operation, after the image fixing operation is performed on one side or the surface of the recording sheet, the switching member of the switchback device **36** switches the direction of conveyance of the recording sheet. Specifically, the direction of conveyance of the recording sheet is switched to a path to a transfer reversal device by the switching member. When the recording sheet is conveyed to the transfer reversal device, the recording sheet is reversed to enter the secondary transfer nip region of the image forming apparatus **100** again. In the image forming apparatus **100**, a toner image is secondarily transferred onto the other side or a back face of the recording sheet so that the secondary transfer process and the fixing process are executed. Then, the recording sheet is ejected onto the ejection tray.

[0076] The scanner **150** fixed to the image forming device **1** includes a fixed reading unit **321** and a movable reading unit **152**. The scanner **150** and an automatic document feeder **51** (simply referred to as an ADF **51**) include fixed reading units. The movable reading unit **152** is disposed

immediately below a second exposure glass **155** (to be described below) that is fixed to the upper wall of a casing of the scanner **150** so as to contact an original document MS. The movable reading unit **152** includes a light source and optical process units such as multiple reflection mirrors, so that these optical units can move in a left and right direction in FIG. **1**. In the course of moving the optical components from left to right in FIG. **1**, the light source emits the light. After the surface of the original document MS placed on the second exposure glass **155** reflects the light, the reflected light is further reflected on multiple reflection mirrors until an image reading sensor **153** that is fixed to the scanner **150** receives the reflected light.

[0077] In contrast, the fixed reading unit includes the first fixed reading unit **151** disposed inside the scanner **150**, and a second fixed reading unit **95** (see FIG. **6**) disposed inside the ADF **51**. The first fixed reading unit **151** includes a light source, multiple reflection mirrors, and the image reading sensor **153** such as CCD. The first fixed reading unit **151** is disposed immediately below a first exposure glass **154** (described below) that is fixed to the upper wall of the casing of the scanner **150**, so that the first exposure glass **154** contacts the original document MS. When the sheet-like original document MS that is conveyed by the ADF **51** (described below) passes over the first exposure glass **154**, the light source emits light. After a document face of the original document MS sequentially reflects the light emitted from the light source, the reflected light is further reflected on multiple reflection mirrors until the image reading sensor **153** receives the reflected light. By so doing, the first face of the original document MS is scanned without moving the optical components such as the light source and the multiple reflection mirrors. Further, the second fixed reading unit **95** scans or reads the second side or the back surface of the original document MS after the original document MS has passed the first fixed reading unit **151**. The original document MS is a sheet-like member such as paper or film.

[0078] FIG. **4** is a perspective view of the scanner **150** and the ADF **51** according to an embodiment of the present disclosure.

[0079] FIG. **5** is a perspective view of the ADF **51** according to an embodiment of the present disclosure.

[0080] FIG. **6** is a schematic diagram illustrating the configuration of the ADF **51** according to an embodiment of the present disclosure.

[0081] The ADF **51** is disposed above the scanner **150**. As illustrated in FIGS. **4** to **6**, the ADF **51** includes a document stacker **53a** on which the document MS before reading is placed, and a document conveyor **54** that conveys the original document MS placed on the document stacker **53a** along the document conveyance path **203**. The document stacker **53a** may be a document loading table.

[0082] The ADF **51** may include a document stacking table **55** for stacking the original documents MS after reading. The ADF **51** may include, specifically, a body cover **52**. The body cover **52** may include the document stacker **53a** and the document conveyor **54**.

[0083] As illustrated in FIG. **4**, the ADF **51** is openably supported by hinges **159** each being fixed to the scanner **150**. The ADF **51** rotatably opens and closes in the upward and downward directions. With the rotation of the ADF **51** in the upward and downward directions, the ADF **51** works as an opening door, so that the first exposure glass **154** and the second exposure glass **155** on the upper face of the scanner **150** are exposed while the ADF **51** is open. In a case of the one-sided bound documents such as a book of a document bundle bounded on one-side, the original documents MS are not separated one by one. For this reason, the original documents MS in the above-described form are not conveyed by the ADF **51**.

[0084] When reading the one-sided bound documents, the ADF **51** is opened as illustrated in FIG. **4**. After the ADF **51** is opened as illustrated in FIG. **4**, the one-sided bound documents are placed on the second exposure glass **155** with a page to be read facing down. Then, the ADF **51** is closed. Then, the scanner **150** causes the movable reading unit **152** to read the image on the page of the one-sided bound documents placed facedown.

[0085] On the other hand, when the original documents MS are in a form of a document bundle of simply accumulated individual original documents MS, the original documents MS are sequentially read by the first fixed reading unit **151** in the scanner **150** or the second fixed reading unit **95** in the ADF **51** while the ADF **51** automatically conveys the original documents MS one by one. In this case, a copy start button **158** is pressed after the bundle of original documents is placed on the document stacker **53a** of the ADF **51**. Then, the ADF **51** starts conveyance of the original documents MS that are a bundle of original documents stacked on the document stacker **53a** to convey the original documents MS sequentially from top of the bundle of original documents MS to the document conveyor **54**, and further convey the original documents MS to the document stacking table **55** while reversing the original document MS. In the course of this conveyance of the original documents MS, immediately after the original document MS is reversed, the original document MS is caused to pass immediately above the first fixed reading unit **151** of the scanner **150**. At this time, the image on the first face of the original document MS is read by the first fixed reading unit **151** of the scanner **150**.

[0086] As illustrated in FIG. **6**, the ADF **51** according to the present embodiment includes a document setting part A, a document separating and feeding part B, a registration part C, a document turning part D, a first reading and conveying part E, a second reading and conveying part F, a document ejecting part G, and a document stacking part H. Further, the ADF **51** includes a document conveyance path **203** for conveying the original document MS from the document stacker **53a** toward the first fixed reading unit **151** that is an image reading position. In other words, the document conveyor **54** includes the document setting part A, the document separating and feeding part B, the registration part C, the document turning part D, the first reading and conveying part E, the second reading and conveying part F, the document ejecting part G, and the document conveyance path **203**.

[0087] The document setting part A has the document stacker **53a** on which a bundle of original documents MS is placed. The document separating and feeding part B separates and feeds the original document MS one by one from the bundle of the original documents MS set on the document stacker **53a**. In the registration part C, the original document MS fed from the document separating and conveying part B temporarily contacts a pullout drive roller **86** and a pullout driven roller **87** to be aligned and fed again. The document turning part D has a conveyance passage curved in a C-shape, and turns the original document MS to be conveyed in the curved conveyance passage so as to reverse the original document MS upside down while turning the original document MS.

[0088] Then, in the first reading and conveying part E, the first fixed reading unit **151** disposed in the scanner **150** below the first exposure glass **154** as illustrated in FIG. **1** reads the first face of the original document MS while the original document MS is being conveyed on the first exposure glass **154**. Further, in the second reading and conveying part F, the second fixed reading unit **95** reads the second face of the original document MS while the original document MS is conveyed under the second fixed reading unit **95**. After the images on both sides of the original document MS are read, the original document MS is conveyed in the document ejecting part G to be ejected toward the document stacking part H. In the document stacking part H, the original documents MS are placed and stacked on the document stacking table **55**.

[0089] The original document MS is placed on the movable document table **53b** in the document setting part A, with the leading end of the original document MS placed on the movable document table **53b** that is pivotable in the direction indicated by arrows "a" and "b" in FIG. **6** depending on the thicknesses of a bundle of the original documents MS and the trailing end of the original document MS placed on the document stacker **53a**. At this time, the side guides of the document stacker **53a** contact both lateral side ends of the original document MS in the width direction (i.e., the direction orthogonal to the drawing sheet) to adjust the position of the original document MS in the width direction. The original documents MS thus set push up a lever **62** that is pivotably

disposed above the movable document table **53b**. Along with this movement of the original documents MS, the document set sensor **63** detects the setting of the original documents MS, and transmits the detection signal to the ADF controller **904** (see FIG. 25). The detection signal is then transmitted from the ADF controller **904** to a scanner controller **903** (see FIG. 25) of the scanner **150** via an interface (I/F).

[0090] The first length sensor **57** and the second length sensor **58** are held on the document stacker **53a**. Each of the first length sensor **57** and the second length sensor **58** includes a reflective photosensor or an actuator-type sensor for detecting the length of the original document MS in the sheet conveyance direction. The length of the original document MS in the sheet conveyance direction is detected with the first length sensor **57** and the second length sensor **58**.

[0091] The pickup roller **80** is supported by the cam mechanism to be movable in the vertical direction (i.e., the directions indicated by arrows “c” and “d” in FIG. 6) and is disposed above the bundle of original documents MS placed on the movable document table **53b**. The cam mechanism is driven by a pickup motor to move the pickup roller **80** in the vertical direction. As the pickup roller **80** moves upward, the movable document table **53b** rotates in the direction indicated by arrow “a” in FIG. 6, so that the pickup roller **80** is brought to contact the uppermost original document MS placed on top of the bundle of original documents MS. As the movable document table **53b** further moves upward, a table lifting sensor **59** detects that the movable document table **53b** moves up to the maximum height. In response to this detection, the pickup motor stops driving to stop the movable document table **53b** from moving up.

[0092] The apparatus control panel including, for example, a numeric keypad and a display included in the image forming apparatus **100** is operated by an operator to perform a key operation for setting a reading mode indicating a double-sided reading mode or a single-sided reading mode and a pressing operation of a copy start key. As the copy start button **158** is pressed by an operator, the document feeding signal is sent from a device controller **901** of the image forming device **1** to the ADF controller **904** of the ADF **51**. In response to the document feeding signal, the pickup roller **80** is rotated along with the forward rotation of the sheet feeding motor **191**, so that the original documents MS on the movable document table **53b** are fed from the movable document table **53b**.

[0093] The setting of the double-sided reading mode or the single-sided reading mode collectively covers the whole original documents MS stacked on the movable document table **53b**. To be more specific, when the double-sided reading mode or the single-sided reading mode is set, both sides or a single-side of the whole original documents MS stacked on the movable document table **53b** can be read.

[0094] In addition, individual reading mode setting can be performed on separate ones of the original documents MS. For example, the double-sided reading mode can be applied to the first and 10th original documents MS while the single-sided reading mode can be applied to the other original documents MS.

[0095] The original document MS fed by the pickup roller **80** enters the document separating and feeding part B to be fed to the contact position with the sheet feed belt **84**. The sheet feed belt **84** is wound and stretched by, for example, a drive roller **82** to be endlessly moved in the clockwise direction in FIG. 5 by rotation of the drive roller **82** along with the forward rotation of the sheet feeding motor **191**. A separation roller **85** is in contact with the lower stretched face of the sheet feed belt **84** to be rotated in the clockwise direction in FIG. 5 along with the forward rotation of the sheet feeding motor **191**. At the contact portion, the sheet feed belt **84** is rotated so that the surface of the sheet feed belt **84** moves in the sheet conveyance direction. By contrast, the separation roller **85** is in contact with the sheet feed belt **84** with a given pressure. When the separation roller **85** directly contacts the sheet feed belt **84** or a single original document MS is nipped in the contact portion, the separation roller **85** is rotated with rotation of the sheet feed belt **84** or movement of the original document MS.

[0096] However, when multiple original documents MS are nipped in the contact portion, the force of the separation roller **85** to be rotated with rotation of the sheet feed belt **84** or movement of the original document MS is lower than the torque of a torque limiter. For this reason, the separation roller **85** is rotated in the clockwise direction that is opposite to a direction in which the separation roller **85** is rotated. As a result, the separation roller **85** applies the force of movement in the direction opposite to the sheet conveyance direction, to the original documents MS under the uppermost original document MS, so that the uppermost original document MS along is separated from the multiple original documents MS under the uppermost original document MS. The above-described operation is referred to as a sheet feeding and separating operation.

[0097] The original document MS is separated from the other original documents MS through the operations of the sheet feed belt **84** and the separation roller **85**, and enters the registration part C. Then, the leading end of the original document MS is detected when the original document MS passes directly under a document contact sensor **72**. At this time, the pickup roller **80** receiving the driving force of the pickup motor is still rotating. However, as the pickup roller **80** is separated from the original document MS due to descendance of the movable document table **53b**, the original document MS is conveyed only by an endless moving force of the sheet feed belt **84**. Then, the endless movement of the sheet feed belt **84** is continued for a given time from the timing at which the leading end of the original document MS is detected by the document contact sensor **72**. Then, the leading end of the original document MS contacts the contact portion of the pullout driven roller **87** and the pullout drive roller **86** that rotates while contacting the pullout drive roller **86**.

[0098] The pullout driven roller **87** has a function of conveying the original document MS to the intermediate roller pair **66** downstream from the pullout driven roller **87** in the document conveyance direction, and is driven and rotated by the rotation of the sheet feeding motor **191** in the reverse direction. As the sheet feeding motor **191** rotates in the reverse direction, the pullout driven roller **87** and one roller of the intermediate roller pair **66** contacting the pullout driven roller **87** start rotating and the endless movement of the sheet feed belt **84** stops. At this time, the pickup roller **80** also stops rotating.

[0099] The original document MS that is fed by the pullout driven roller **87** passes directly under the document width sensor **73**. The document width sensor **73** includes multiple document detectors each including a reflective photosensor. The multiple document detectors are aligned in a row in the width direction of the original document MS (i.e., the direction perpendicular to the drawing sheet of FIG. **6**). The size of the original document MS in the width direction is detected based on which one of the multiple document detectors detects the original document MS. The length of the original document MS in the document conveyance direction is detected based on the time from when the leading end of the original document MS is detected by the document contact sensor **72** to when the trailing end of the original document MS is not detected by the document contact sensor **72**.

[0100] The leading end of the original document MS whose size in the width direction is detected by the document width sensor **73** enters the document turning part D and is nipped by the contact portion between the rollers of the intermediate roller pair **66**. The conveyance speed of the original document MS conveyed by the intermediate roller pair **66** is set faster than the conveyance speed of the original document MS in the first reading and conveying part E that will be described below. This configuration achieves a reduction in time for conveying the original document MS to the first reading and conveying part E.

[0101] The leading end of the original document MS conveyed in the document turning part D passes through a position where the leading end of the original document MS faces the scan entrance sensor **67**. As a result, when the leading end of the original document MS is detected by the scan entrance sensor **67**, the document conveyance speed of the original document MS by the intermediate roller pair **66** is reduced until the leading end of the original document MS is

conveyed to the position of the scan entrance roller pair (including rollers **89** and **90**) downstream from the scan entrance sensor **67** in the sheet conveyance direction. As a document reading motor starts to drive and rotate, one roller of the scan entrance roller pair (including the rollers **89** and **90**), one roller of a first scan exit roller pair **92**, and one roller of a second scan exit roller pair **93** respectively start rotation.

[0102] In the document turning part D, while the original document MS is conveyed in the curved conveyance passage between the intermediate roller pair **66** and the scan entrance roller pair (including the rollers **89** and **90**), the upper and lower faces of the original document MS are reversed, and the conveyance direction of the original document MS is turned back. Then, the leading end of the original document MS that has passed through the nip region between the rollers (**89** and **90**) of the scan entrance roller pair passes directly under the registration sensor **65**. The operation up to this point after the sheet feeding and separating operation is referred to as a “document pullout operation”.

[0103] When the registration sensor **65** detects the leading end of the original document MS, the conveyance speed of the original document MS is gradually decreased through the given conveyance distance. Then, the rotation of the pullout drive roller **86** and the intermediate roller pair **66** is stopped due to the stop of the sheet conveyance motor **192**, which will be described below, and the rotation of the scan entrance roller pair is stopped due to the stop of the document reading motor. Due to this action, the conveyance of the original document MS is temporarily stopped at the registration position before the first reading and conveying part E. Further, a registration stop signal is sent to the scanner controller **903** of the scanner **150**.

[0104] FIG. **7** is a partially enlarged view of the main part of the ADF **51** of FIG. **5**.

[0105] FIG. **8** is a top view of a part of the document conveyance path **203** of the ADF **51** according to an embodiment of the present disclosure.

[0106] FIG. **9** is a perspective view of the sound generators **212** according to an embodiment of the present disclosure.

[0107] FIG. **10** is a perspective view of the sound generator **212** according to a first modification of the sound generators **212** of an embodiment of the present disclosure.

[0108] As illustrated in FIGS. **7** to **10**, the ADF **51** includes one or more sound generators **212** on the document conveyance path **203** and having a protrusion or a recess with respect to the surface of the document conveyance path **203**, and sound collectors **201** that perform sound collection when any one of the sound generators **212** and a fastening object attached to the original document MS come into contact with each other.” The sound collection indicates collecting sound.

[0109] As illustrated in FIG. **8**, the sound generators **212** are continuously or intermittently disposed from one end to the other end in the width direction of the document conveyance path **203**. When the original document MS with a fastening object such as a clip or a stapler is conveyed to the document conveyance path **203**, the sound generators **212** and the fastening object come into contact with each other, and abnormal sound different from normal operation sound such as a sound of collision or a sound of rubbing when the fastening object rides over the sound generators **212** is generated. The noise generated by the fastening object is referred to as “abnormal sound”. Since the sound generators **212** are disposed over the entire width direction of the document conveyance path **203**, the fastening object and the sound generators **212** can be brought into contact with each other to generate an abnormal sound regardless of the position of the fastening object in the width direction of the document conveyance path **203**.

[0110] The ADF **51** includes a prediction unit **202** that predicts whether any fastening object is attached to an original document MS or not, based on the sound collected by the sound collectors **201**. When any of the sound generators **212** and the fastening object come into contact with each other, the abnormal sound is mixed with the operation sound collected by the sound collectors **201**, and the operation sound becomes a characteristic operation sound, which can be distinguished from the operation sound during normal conveyance. For this reason, the prediction unit **202** can predict

the presence or absence of the fastening object based on the sound collected by the sound collectors **201**.

[0111] As described above, the fastening object attached to the original document MS can be detected by the ADF **51**.

[0112] FIG. **11** is a schematic diagram illustrating an example of the original document MS to which a fastening object **220** is attached.

[0113] FIG. **12** is a diagram illustrating conveyance of the original document MS of FIG. **11**.

[0114] FIG. **13** is a schematic diagram illustrating another example of the original document MS to which the fastening object **220** is attached.

[0115] FIG. **14** is a diagram illustrating conveyance of the original document MS of FIG. **13**.

[0116] In the example illustrated in FIGS. **11** and **12**, the uppermost document MS is conveyed to the document conveyance path **203** while the fastening object **220** is attached to the original document MS. In such a case, when the fastening object **220** contacts any part of the document conveyance path **203**, the conveyance of the original document MS is stopped, which leads to occurrence of a jam.

[0117] In the example illustrated in FIGS. **13** and **14**, it is conceivable that the fastening object **220** remains at the place together with the document bundle, and the uppermost document MS is pulled out and further conveyed. However, in either case, any one of the original documents MS in the document bundle is finally conveyed together with the fastening object **220**. Since the fastening object **220** is attached to the side face in the document conveyance direction, when the fastening object **220** comes into contact with any part of the document conveyance path **203** during the conveyance of the original document MS, the fastening object **220** is removed from the original document MS and falls onto the document conveyance path **203**. If the subsequent original document is conveyed when the fastening object **220** remains on the document conveyance path **203**, the subsequent original document collides with the fastening object **220** dropped on the document conveyance path **203**, which also causes a jam.

[0118] With the ADF **51** according to the present embodiment, in the cases as illustrated in FIGS. **11** to **14**, in other words, regardless of the position where the fastening object passes in the document conveyance path **203**, whether the fastening object is attached to the original document MS can be predicted and the fastening object can be detected.

[0119] A description is given of the sound generator **206** having a protrusion on the surface of the document conveyance path **203**.

[0120] As illustrated in FIG. **9**, the sound generator **206** includes a first face **206a** and a sloped face **206b**. The first face **206a** extends from the surface of the document conveyance path **203** upward in the vertical direction. The sloped face **206b** couples to the upper end of the first face **206a** and the surface of the document conveyance path **203**. The sloped face **206b** extends from the upper end of the first face **206a** in the opposite direction from the conveyance direction CD. In other words, the sloped face **206b** is disposed to contact the leading end of the original document MS that is under conveyance. The first face **206a** is a face along in the width direction of the document conveyance path **203**.

[0121] A description is given of the sound generator **207** having a recess in the surface of the document conveyance path **203**.

[0122] As illustrated in FIG. **10**, the sound generator **207** includes a first face **207a** and a sloped face **207b**. The first face **207a** extends from the surface of the document conveyance path **203** downward in the vertical direction. The sloped face **207b** couples to the lower end of the first face **207a** and the surface of the document conveyance path **203**. The sloped face **207b** extends from the lower end of the first face **207a** in the conveyance direction CD. In other words, the sloped face **207b** is disposed to contact the leading end of the original document MS that is under conveyance. The sound generator **207** may be a groove formed in the surface of the document conveyance path **203**. The first face **207a** is a face along in the width direction of the document conveyance path

203.

[0123] As described above, the sound generators **212** include the first faces **206a** and **207a** extending from the surface of the document conveyance path **203** in the vertical direction, and the sloped faces **206b** and **207b** extending from the ends of the first faces **206a** and **207a** to the surface of the document conveyance path **203** in the vertical direction. It is preferable that the multiple sound generators **212** are disposed at intervals in the conveyance direction CD of the original document MS. With this configuration, the ADF **51** can increase the volume of the abnormal sound and the chances of occurrence of abnormal sound, and can enhance the detection accuracy of a fastening object.

[0124] Since the sound generators **212** have the sloped faces **206b** and **207b** as described above, the original document MS can be conveyed along the sloped faces **206b** and **207b**. For this reason, even if there is a recess or a protrusion on the document conveyance path **203** due to the sound generators **212**, the ADF **51** restrains or prevents conveyance of the original document MS from being hindered by the recess or the protrusion, and can prevent the original document MS from being damaged.

[0125] As illustrated in FIG. **8**, the document conveyance path **203** is provided with the separation roller **85**, the pullout driven roller **87**, and the intermediate roller pair **66**, in this order from the upstream side in the conveyance direction CD. In FIG. **8**, a curved portion between the separation roller **85** and the pullout driven roller **87** on the conveyance path is omitted. Specifically, the sound generators **212** may be disposed on both sides in the width direction of the document conveyance path **203**, interposing the separation roller **85**, the pullout driven roller **87**, and the intermediate roller pair **66**. In addition, in a region where the separation roller **85**, the pullout driven roller **87**, and the intermediate roller pair **66** are not disposed in the width direction of the document conveyance path **203**, the sound generators **212** may be disposed so as to extend from one end to the other end in the width direction of the document conveyance path **203**.

[0126] It is preferable that the sound generators **212** are disposed upstream from the downstream end of the intermediate roller pair **66** in the conveyance direction CD. As illustrated in FIG. **6**, the document conveyance path **203** has a largely curved portion on the downstream side from intermediate roller pair **66** in the conveyance direction CD. When the original document MS with the fastening object attached enters the document conveyance path **203** downstream from the intermediate roller pair **66** in the conveyance direction CD, the fastening object does not pass the largely curved portion and a jam occurs. This may cause a significantly damage on the original document MS, for example, the original document MS may be wrinkled in a bellows shape. Further, in the document conveyance path **203** downstream from the intermediate roller pair **66** in the conveyance direction CD, when the fastening object comes off from the original document MS, the fastening object may fall down from the document conveyance path **203**, which makes the removal of the fastening object difficult.

[0127] Accordingly, with the ADF **51** provided with the sound generators **212** upstream from the downstream end of the intermediate roller pair **66** in the conveyance direction CD, whether the fastening object is attached to the original document MS can be predicted before the original document MS reaches the intermediate roller pair **66**, and the damage on the original document MS can be prevented.

[0128] The sound collectors **201** may be sound collection microphones. The sound collectors **201** are attached to the sheet feeder cover **98**, and may be disposed at least above the document conveyance path **203**. The sound collectors **201** may be disposed, for example, on the back of the sheet feeder cover **98** of the ADF **51**.

[0129] As illustrated in FIGS. **5** to **7**, the sound collectors **201** may include a first sound collector **201a** and a second sound collector **201b**. The first sound collector **201a** is disposed on the back face of the sheet feeder cover **98** upstream from the pickup roller **80** in the conveyance direction CD. The second sound collector **201b** is disposed on the back face of the sheet feeder cover **98**

downstream from the drive roller **82** and upstream from the sheet feeder cover **98** from the pullout drive roller **86** in the conveyance direction CD. The number of sound collectors **201** may be one or more. The positions of the sound collectors **201** are not limited to the positions illustrated in FIGS. 5 to 7.

[0130] FIG. **15** is a perspective view of a sound generator according to a second modification of the sound generators **212** of an embodiment of the present disclosure.

[0131] As illustrated in FIG. **15**, a sound generator **208** may include a contact face **208a** with the original document MS, and a cutout portion **209** in the contact face **208a** or a rough face portion having the surface roughness greater than the surface roughness of the document conveyance path **203**. The surface roughness may be, for example, an arithmetic average roughness (Ra).

[0132] Due to such a configuration, in addition to the sound generated when the fastening object collides with the contact face **208a**, abnormal sound is generated as the fastening object rubs against the contact face **208a** when the fastening object moves on the contact face **208a**. With this configuration, the ADF **51** can increase the chances of occurrence of abnormal sound, and can enhance the detection accuracy of a fastening object. In the example illustrated in FIG. **15**, the sound generator **208** has a protrusion on the surface of the document conveyance path **203**.

However, the same effect is achieved even in a case where the sound generator **207** having a recess has a cut portion or a rough face portion on the sloped face **207b** that is the contact face with the original document MS, as in the example illustrated in FIG. **10**.

[0133] FIG. **16** is a perspective view of a sound generators according to a third modification of the sound generators **212** of an embodiment of the present disclosure.

[0134] In a case where a sound generator **210** has a protrusion on the surface of the document conveyance path **203**, as illustrated in FIG. **16**, the sound generator **210** may include a plate member **211**, a protrusion **213**, and an elastic member **214**. The plate member **211** includes a sloped face **211a**. The protrusion **213** is disposed on the side face of the plate member **211**. The elastic member **214** supports the plate member **211** to be movable in a direction orthogonal to the surface of the document conveyance path **203**. The protrusion **213** may be fixed to, for example, ribs **204** (described below) formed on the document conveyance path **203**. The plate member **211** is rotatable about the protrusion **213**.

[0135] Since the sound generator **210** having the protrusion on the surface of the document conveyance path **203** includes the plate member **211** and the elastic member **214**, when the fastening object passes on the plate member **211**, the plate member **211** is lowered. Then, when the fastening object passed the plate member **211**, the plate member **211** is bounced back upward by the elastic member **214**. At this time, abnormal sound (noise) such as sound generated by the plate member **211** or the elastic member **214** or sound generated when the plate member **211** comes into contact with the original document MS being conveyed is generated. With this configuration, the ADF **51** can increase the chances of occurrence of abnormal sound (noise), and can enhance the detection accuracy of a fastening object.

[0136] FIG. **17** is a perspective view of a sound generator according to a fourth modification of the sound generators **212** of an embodiment of the present disclosure.

[0137] The sound generator **210** may include, in addition to the configuration of FIG. **16**, an opposing member **231** disposed above the plate member **211** so as to face the plate member **211**. With this configuration, when the plate member **211** is bounced back upward by the elastic member **214**, a portion of the sloped face **211a** of the plate member **211**, which is located outside the original document MS in the widthwise direction, collides with the opposing member **231**. By so doing, the volume of the generated abnormal sound increases. Accordingly, the ADF **51** can enhance the detection accuracy of the fastening object.

[0138] The opposing member **231** is rotatably fixed to a top board **230** disposed facing the upper part of the document conveyance path **203**. One end of the opposing member **231** is fixed to the top board **230** and extends toward the plate member **211**. In other words, the opposing member **231** is

suspended from the top board **230** and the other end of the opposing member **231** is movable. The other end of the opposing member **231** is movable along, for example, the conveyance direction CD. Multiple opposing members **231** may be disposed along the width direction of the document conveyance path **203**. The opposing member **231** may have a shape in which a portion on the other end is curved so as to extend toward the downstream side in the conveyance direction CD. The other end of the opposing member **231** is in contact with the plate member **211**. Multiple elastic members **214** may be disposed on the document conveyance path **203**.

[0139] The material of the opposing member **231** may be a metal. With this configuration, the material of the opposing member is metallic. Since the material of the opposing member is metallic, the portion of the sloped face **211a** of the plate member **211** located outside the original document MS in the widthwise direction collides with the opposing member, which can generate the sound having a different frequency from the operation sound. Accordingly, the ADF **51** can enhance the detection accuracy of the fastening object.

[0140] FIG. **18** is a top view of a part of the document conveyance path **203** according to the first modification of an embodiment of the present disclosure.

[0141] FIG. **19** is a diagram illustrating the part of the document conveyance path **203** of FIG. **18**, viewed from the upstream side in the conveyance direction CD.

[0142] FIG. **20** is a side view of the part of the document conveyance path **203** of FIG. **18**.

[0143] As illustrated in FIGS. **18** to **20**, the ADF **51** may include the rib **204** on the document conveyance path **203** and extend in the conveyance direction CD of the original document MS. The ribs **204** have a function of reducing a frictional force when the rib **204** comes into contact with the original document MS during the conveyance of the original document MS.

[0144] Multiple ribs **204** may be disposed at intervals in the width direction of the document conveyance path **203**.

[0145] FIG. **21** is a perspective view of a sound generator according to a fifth modification of the sound generators **212** of an embodiment of the present disclosure.

[0146] As illustrated in FIG. **21**, one or more sound generators **205** may be disposed on the rib **204**. The sound generator **205** has recesses in the surface of the document conveyance path **203**. When the ADF **51** has the rib **204** or the ribs **204**, it is difficult to dispose the sound generators **212** having the protrusion from one end to the other end in the width direction of the document conveyance path **203**. With the sound generator **205** having the recesses in the surface of the rib **204**, abnormal sound is generated when a fastening object passes on the rib **204**. Accordingly, the ADF **51** can detect the fastening object.

[0147] As illustrated in FIG. **21**, the sound generator **205** includes a first face **205a** and a sloped face **205b**. The first face **205a** extends from the surface of the rib **204** downward in the vertical direction. The sloped face **205b** couples to the lower end of the first face **207a** and the upper face of the rib **204**. The sloped face **205b** extends from the lower end of the first face **205a** in the conveyance direction CD. In other words, the sloped face **205b** is disposed to contact the leading end of the original document MS that is under conveyance. The sound generator **205** may be a groove formed in the upper face of the rib **204**. The first face **205a** is a face along the width direction of the document conveyance path **203**. The sound generator **205** may be disposed on each of the multiple ribs **204** and arranged on the same straight line in the width direction of the document conveyance path **203**.

[0148] FIG. **22** is a top view of a part of the document conveyance path **203** according to the second modification of an embodiment of the present disclosure.

[0149] FIG. **23** is an enlarged perspective view of a part of the document conveyance path **203** of FIG. **22**.

[0150] When the multiple ribs **204** are disposed spaced apart from each other in the width direction of the document conveyance path **203**, the multiple sound generators **212** may have a protrusion on the document conveyance path **203** and be disposed to fill the spaces between the adjacent ribs **204**

in the width direction of the document conveyance path **203**, as illustrated in FIGS. **22** and **23**. Further, the height of the sound generators **212** on the surface of the document conveyance path **203** is preferably lower than the height of the ribs **204** on the surface of the document conveyance path **203**.

[0151] When the height of the sound generators **212** on the surface of the document conveyance path **203** is equal to or greater than the height of the ribs **204** on the surface of the document conveyance path **203**, the original document MS is conveyed while contacting the sound generators **212**. Due to such a configuration, the original document MS may be damaged. Since the height of the sound generators **212** on the surface of the document conveyance path **203** is lower than the height of the ribs **204** on the surface of the document conveyance path **203**, the probability that the original document MS contacts the sound generators **212** can be reduced, and damage on the original document MS can be restrained or prevented. In addition, since the probability that the original document MS contacts the sound generators **212** is reduced, it is easier to detect abnormal sound generated when the fastening object contacts the sound generators **212**. Accordingly, the ADF **51** can enhance the detection accuracy of the fastening object.

[0152] The difference between the height of the sound generators **212** on the surface of the document conveyance path **203** and the height of the ribs **204** on the surface of the document conveyance path **203** can be set to be equal to or less than, for example, the thickness of the fastening object.

[0153] In the example illustrated in FIGS. **22** and **23**, the upper face of the ribs **204** is a flat face. However, the configuration is not limited to the above-described configuration. For example, the sound generator **205** illustrated in FIG. **21** may be disposed on the upper face of the ribs **204**. This configuration can increase the abnormal sound generated when the fastening object contacts the sound generators **212**.

[0154] FIG. **24** is a top view of a part of the document conveyance path **203** of the ADF **51** according to another example of an embodiment of the present disclosure.

[0155] As illustrated in FIG. **24**, the multiple sound generators **212** may be arranged at different intervals in the conveyance direction CD of the original document MS. With this configuration, when the fastening object passes over the sound generators **212**, a sound, particularly a collision sound, irregularly, can be generated, a sound of a specific rhythm can be generated. With this configuration, the prediction unit **202** can predict whether a fastening object is attached to the original document MS based on the specific rhythm, and the ADF **51** can enhance the detection accuracy of the fastening object.

[0156] Examples of the material of the surface of the sound generators **212** include, for example, resin and metal. The material of the surface of the sound generators **212** is preferably metal. Accordingly, when the fastening object includes metal, the frequency of the abnormal sound generated due to the contact of the fastening object and the sound generators **212** and the frequency of the normal operating sound increases. Accordingly, the ADF **51** can enhance the detection accuracy of the fastening object.

[0157] FIG. **25** is a block diagram illustrating a part of an electric circuit of the image forming apparatus **100** according to an embodiment of the present disclosure.

[0158] The electric circuit of the image forming apparatus **100** includes, for example, the device controller **901** of the image forming device **1**, the scanner controller **903** of the scanner **150**, and the ADF controller **904** of the ADF **51**. These controllers (i.e., the device controller **901**, the scanner controller **903**, and the ADF controller **904**) include, for example, a central processing unit (CPU), a random access memory (RAM), and a read-only memory (ROM). The device controller **901** and the scanner controller **903** are connected to communicate with each other. The scanner controller **903** and the ADF controller **904** are connected to communicate with each other. The ADF controller **904** and the device controller **901** are connected to communicate with each other.

[0159] The ADF controller **904** is connected to a registration sensor **65**, a document set sensor **63**, a

document ejection sensor **61**, the document contact sensor **72**, a document width sensor **73**, a scan entrance sensor **67**, the first length sensor **57**, and the second length sensor **58**, as illustrated in FIG. **25**. The ADF controller **904** is further connected to, for example, a sheet feeding motor **191**, a sheet conveyance motor **192**, a pullout clutch **193**, a sheet ejection clutch **194**, and a prediction unit **202**. The sound collectors **201** are connected to the prediction unit **202**.

[0160] The sheet conveyance motor **192** connected to the ADF controller **904** is a rotation drive source for the pullout drive roller **86** and the document ejection roller pair **94** in the ADF **51**. The pullout clutch **193** connected to the ADF controller **904** connects and disconnects the rotation driving force of the sheet conveyance motor **192** to and from the pullout drive roller **86**. Then, the sheet ejection clutch **194** connects and disconnects the rotation drive force of the sheet conveyance motor **192** to and from the document ejection roller pair **94** as a feeding and conveying device.

[0161] After receiving the registration stop signal from the ADF controller **904**, the scanner controller **903** sends the reading start signal as a sheet feeding permission signal to the ADF controller **904**. Then, the ADF controller **904** restarts rotating the sheet conveyance motor **192** and the document reading motor. Then, the timing at which the leading end of the original document MS reaches the reading position of the first fixed reading unit **151** is calculated based on the pulse counts of the document reading motor. At this timing, the ADF controller **904** sends a gate signal indicating an effective image area of the first face of the original document MS in the sub-scanning direction, to the scanner controller **903**. The ADF controller **904** continues sending the gate signal to the scanner controller **903** until the trailing end of the original document MS passes through the reading position of the first fixed reading unit **151**, so that the first face of the original document MS is scanned by the first fixed reading unit **151**.

[0162] The original document MS that has passed through the first reading and conveying part E passes through the first scan exit roller pair **92**, which will be described below. Then, the leading end of the original document MS is detected by the document ejection sensor **61**. When the single-sided reading mode is set, the second face of the original document MS is not to be read by the second fixed reading unit **95**, which will be described below. Then, when the leading end of the original document MS is detected by the document ejection sensor **61**, the sheet ejection clutch **194** connects the driving force of the conveyance motor to the document ejection roller pair **94**. The timing at which the trailing end of the original document MS passes through the nip region of the document ejection roller pair **94** is calculated based on the pulse counts of the document ejection motor after the detection of the leading end of the original document MS by the document ejection sensor **61**. Then, based on this calculation result, the sheet ejection clutch **194** is stopped.

[0163] On the other hand, when the double-sided reading mode is set, the document ejection sensor **61** initially detects the leading end of the original document MS. Then, the timing of the period to which the original document MS reaches the second fixed reading unit **95** is calculated based on the pulse counts of the document reading motor. Then, at this timing, the ADF controller **904** sends a gate signal indicating the effective image area of the second face of the original document MS in the sub-scanning direction, to the scanner controller **903**. The ADF controller **904** continues sending the gate signal to the scanner controller **903** until the trailing end of the original document MS passes through the reading position of the second fixed reading unit **95**, so that the second face of the original document MS is scanned by the second fixed reading unit **95**.

[0164] The second fixed reading unit **95** includes contact-type image sensors (CIS) and is coated on the reading face for the purpose of preventing reading vertical streaks due to the paste-like foreign substance adhering to the original document MS to adhere to the reading face of the second fixed reading unit **95**. A second reading roller **96** as a document supporter that supports the original document MS from a non-reading face is disposed at a position facing the second fixed reading unit **95**. The second reading roller **96** functions as a floating retainer that prevents the original document MS from floating up at the reading position by the second fixed reading unit **95** and as a reference white portion for acquiring in the shading data on the second fixed reading unit **95**.

[0165] The prediction unit **202** can predict whether a fastening object is attached to the original document MS, using parameters acquired by performing machine learning in advance. The description of machine learning will be given below.

[0166] FIG. **26** is a flowchart of a fastening object determination process executed by the ADF controller **904** according to an embodiment of the present disclosure.

[0167] The ADF controller **904** separates the original documents MS one by one from the bundle of original documents MS set on the document stacker **53a** in the document setting part A to feed the separated original document MS. Simultaneously, the ADF controller **904** executes the process following the flow of the flowchart illustrated in FIG. **26**, on each of the original documents MS. In the present embodiment, the sheet feeding operation of the original documents MS is continuously performed with given intervals between adjacent original documents MS to such an extent that the original documents MS do not overlap with each other in order to increase the throughput of reading. For this reason, multiple original documents MS are simultaneously present at the intervals between adjacent original documents MS in the document conveyance path **203**.

Accordingly, the flow of the flowchart of FIG. **26** is executed in parallel with an interval for each of the original documents MS.

[0168] When the reading instruction is transmitted from the device controller **901** to the ADF controller **904**, the ADF controller **904** starts the document conveyance operation (step S1). Simultaneously, the ADF controller **904** starts collecting the operation sound by the sound collectors **201** (step S2). Specifically, in step S1, the ADF controller **904** starts driving the pickup roller **80**. Then, the original document MS enters the document conveyance path **203**, and passes over the sound generators **212**. When the original document MS has the fastening object, abnormal sound is generated. The prediction unit **202** acquires the collected operation sounds from the sound collectors **201**, and starts the prediction process of abnormal conveyance, in other words, the prediction process of whether the fastening object is attached to the original document MS, based on the operation sound (step S3).

[0169] In step S3, the prediction process of whether a fastening object is attached to the original document MS is executed at a constant cycle. In step S3, the prediction unit **202** determines whether the original document MS is normally conveyed or abnormally conveyed (with or without a fastening object) using “parameters acquired in advance by machine learning”.

[0170] The “parameter obtained by machine learning in advance” can be generated as follows. The sound of the passing of the original document MS without the fastening object and the sound of the passing of the original document MS with the fastening object are recorded under various conditions, in other words, by varying the error factors. Then, the sound is labeled as “with fastening object” or “without fastening object” to be subjected to machine learning. The generated parameters correspond to a “trained model” in the description of machine learning to be described below.

[0171] A description is now given of an outline of machine learning.

[0172] Machine learning is one of methods for analyzing data, and is a method in which a computer (machine) performs “automatic learning” from datasets to find rules or patterns in the background of the datasets. The rules or patterns obtained as a result of learning is called a “trained model”. Prediction and determination can be performed by using this trained model obtained (learned) by machine learning in advance.

[0173] Examples of the machine learning include a Mahalanobis Taguchi (MT) method and a support vector machine. The following description is given of the MT method. The audio signals collected by the sound collectors **201** are temporarily stored in the RAM of the ADF controller **904**. The short-time Fourier transform (STFT) is performed on the audio signals stored in the RAM of the ADF controller **904** by dividing a longer time signal into shorter segments of equal length (frame processing), and then computing the Fourier transform separately on each shorter segment. By so doing, the time sequence of the power spectrum of the audio signal can be calculated in real

time. The ADF controller **904** then performs a characterization process sequentially on the time sequence of the calculated power spectrum to calculate the feature amount of the sound. The feature amount of the sound is also stored in the RAM of the ADF controller **904**. The characterization process may include, for example, time integration of power in a given frequency band and spectral flux between successive frames.

[0174] Subsequently, the abnormality in conveyance is predicted by using the accumulated feature amount of the sound as an input. A method of predicting the abnormal conveyance is to use the Mahalanobis distance by the MT method. In the field of quality engineering, the MT method is one of Mahalanobis-Taguchi (MT) system known for prediction, diagnosis and analysis based on multidimensional information data. The MT system is a general term for methods for analyzing multidimensional information data, such as the MT method, the MTA method, the TS method, the T method (1), the T method (2), the RT method, and the mean-squared deviation method. The MT method is a method using the Mahalanobis distance, and is a simple and relatively accurate method in which the correlation between items is considered.

[0175] An inverse matrix R^{-1} of the correlation matrix of a unit space dataset used when calculating the Mahalanobis distance is stored in the ROM of the ADF controller **904**. The inverse matrix R^{-1} of the correlation matrix is obtained in advance by creating a unit space dataset (i.e., reference dataset) based on a feature amount obtained from a sound signal by reading a normal original document MS under various erroneous factor conditions in the ADF **51**, and is stored in the ROM of the ADF controller **904**. The obtained unit space dataset corresponds to a “trained model”.

[0176] In a case where the Mahalanobis distance is equal to or smaller than the threshold value Th , it is determined that the conveyance is normal (without fastening object). On the other hand, in a case where the Mahalanobis distance is greater than the threshold value Th , it is determined that the conveyance is abnormal (with fastening object). The threshold value Th is set to a value that can be determined as normal with a margin with respect to a unit space dataset (i.e., a reference dataset) created under various erroneous factor conditions in the ADF **51**. Accordingly, the threshold value Th is set so as to hardly cause false positives, instead of allowing false negatives to some extent. In step **S3**, the prediction process of whether a fastening object is attached to the original document MS is executed using the machine learning as described above.

[0177] The prediction unit **202** determines whether the conveyance is normal or abnormal (whether a fastening object is attached to the original document MS) (step **S4**). When the conveyance is abnormal (with fastening object) (YES in the step **S4**), the prediction process (prediction of abnormality in conveyance) is terminated (step **S5**). At this time, sound collection by the sound collectors **201** is also terminated. After step **S5**, the drive of the sheet feeding motor **191** is stopped, and the pickup roller **80** is elevated to stop the document conveying operation (step **S6**). At this point, the reading operation of the original document MS is not performed. After step **S6**, the ADF controller **904** transmits the result indicating that the conveyance is abnormal, to the device controller **901** (step **S7**).

[0178] When the device controller **901** receives the result indicating that there is abnormal conveyance from the ADF controller **904**, the device controller **901** completes the reading instruction. The device controller **901** displays a message “Abnormal conveyance of original document is detected. Please check original document” on the display of the apparatus operation unit **902** with an abnormal conveyance alarm sound, and notifies the user of the conveyance state.

[0179] On the other hand, when the prediction unit **202** determines that the conveyance is normal (without fastening object) (No in step **S4**), the document conveyance operation is continued (step **S8**). After step **S8**, the ADF controller **904** determines whether the original document MS has passed by a certain distance from the document contact sensor **72** (step **S9**). The certain distance from the document contact sensor **72** may be, for example, a distance from the document contact sensor **72** to the intermediate roller pair **66**. When the ADF controller **904** determines that the original document MS has passed by a certain distance from the document contact sensor **72** (YES

in step S9), the prediction process of abnormal conveyance ends (step S10). At this time, sound collection by the sound collectors **201** is also terminated.

[0180] After step S10, the ADF controller **904** continues the document conveyance operation (step S11), and starts the process for reading and conveying the original document MS (step S12). In addition, between step S11 and step S12, the process of detecting abnormal conveyance may be performed by another sensor that is provided in the related art. After step S12, the process for reading and conveying the original document MS is terminated (step S13).

[0181] After step S13, the ADF controller **904** transmits a result indicating that there is no abnormal conveyance, to the device controller **901** (step S7). As the device controller **901** receives the result indicating that there is no abnormal conveyance, from the ADF controller **904**, the device controller **901** completes the reading instruction.

[0182] When the ADF controller **904** determines that the original document MS has not passed a certain distance from the document contact sensor **72** (NO in step S9), the process goes back to step S4 and the prediction unit **202** determines again whether there is any abnormal conveyance (whether there is a fastening object attached to the original document MS).

[0183] The image forming apparatus **100** according to the present embodiment includes an automatic document feeder **51** and an image forming device **1**. With this configuration, the image forming apparatus **100** includes the automatic document feeder **51**, and thus can detect a fastening object attached to the original document MS.

ASPECTS OF THE PRESENT DISCLOSURE

[0184] A description is given below of some aspects of the present disclosure.

Aspect 1

[0185] In Aspect 1, an automatic document feeder includes a document stacker on which a document is placed, a document conveyor to convey the document placed on the document stacker, along a document conveyance path, a sound generator disposed on the document conveyance path and having a protruding shape or a recessed shape to a plane of the document conveyance path, a sound collector to collect sound when the sound generator and a fastening object attached to the document contact each other, and a prediction unit to predict whether the fastening object is attached to the document, based on the sound collected by the sound collector. Multiple sound generators including the sound generator are disposed serially or intermittently from one end to an opposite end in the width direction of the document conveyance path.

Aspect 2

[0186] In Aspect 2, in the automatic document feeder according to Aspect 1, each of the multiple sound generators includes a first face that extends from the surface of the document conveyance path in the vertical direction, and a sloped face extending from an end of the first face to the surface of the document conveyance path in the vertical direction. The multiple sound generators are disposed at intervals in a conveyance direction of the document.

Aspect 3

[0187] In Aspect 3, the automatic document feeder according to Aspect 1 further includes a rib on the document conveyance path and extending along the conveyance direction of the document. Each of the multiple sound generators has a recess in the surface of the document conveyance path. At least one of the multiple sound generators is disposed on a top face of the rib.

[0188] Aspect 4 In Aspect 4, in the automatic document feeder according to Aspect 1, the sound generator having the protrusion includes a plate member having a sloped face, and an elastic member supporting the plate member to be movable in a direction orthogonal to the surface of the document conveyance path.

Aspect 5

[0189] In Aspect 5, in the automatic document feeder according to any one of Aspects 1 to 4, the sound generator has a cutout portion in a contact face with the document or a rough surface that has a surface roughness greater than the surface roughness of the document conveyance path.

Aspect 6

[0190] In Aspect 6, the automatic document feeder according to Aspect 1 or Aspect 2 further includes ribs disposed on the document conveyance path and extending in a conveyance direction of the document. The ribs are disposed at intervals in the width direction of the document conveyance path. Each of the multiple sound generators has a protrusion protruding toward the surface of the document conveyance path. The multiple sound generators are disposed to fill a gap between adjacent ribs in the width direction of the document conveyance path. The height of the sound generator to the surface of the document conveyance path is lower than the surface of the document conveyance path.

Aspect 7

[0191] In Aspect 7, the automatic document feeder according to Aspect 4 includes an opposing member disposed facing the plate member above the plate member.

Aspect 8

[0192] In Aspect 8, in the automatic document feeder according to Aspect 2, the multiple sound generators are disposed at different intervals in a conveyance direction of the document.

Aspect 9

[0193] In Aspect 9, an image forming apparatus includes the automatic document feeder according to any one of Aspects 1 to 8, and an image forming device to form an image on a medium.

Aspect 10

[0194] In Aspect 10, an automatic document feeder includes a document stacker, a document conveyor, a sound generator, a sound collector, and circuitry. The document stacker is a receptacle on which a document is placed. The document conveyor conveys the document on the document stacker along a document conveyance path in a conveyance direction. The sound generator is disposed on a surface of the document conveyance path, the sound generator having a protrusion or a recess protruding or recessed from the surface of the document conveyance path and extending continuously or intermittently from one end to another end of the document conveyance path in a width direction orthogonal to the conveyance direction. The sound collector collects sound generated by contact between the sound generator and a fastening object attached to the document. The circuitry is to predict whether the fastening object is attached to the document, based on the sound collected by the sound collector.

Aspect 11

[0195] In Aspect 11, the automatic document feeder according to Aspect 10 further includes multiple sound generators including the sound generator. The multiple sound generators are disposed at intervals in the conveyance direction. Each of the multiple sound generators includes a first face and a second face. The first face is raised vertically from the surface of the document conveyance path. The second face is inclined relative to the first face and connecting an edge of the first face and the surface of the document conveyance path.

Aspect 12

[0196] In Aspect 12, in the automatic document feeder according to Aspect 11, the multiple sound generators are disposed at different intervals in the conveyance direction.

Aspect 13

[0197] In Aspect 13, the automatic document feeder according to Aspect 10 further includes a rib on the document conveyance path and extending in the conveyance direction. The sound generator has the recess recessed from a surface of the rib.

Aspect 14

[0198] In Aspect 14, in the automatic document feeder according to any one of Aspects 10 to 13, the protrusion of the sound generator includes, a plate having a sloped face, and an elastic member movably supporting the plate in a direction orthogonal to the surface of the document conveyance path.

Aspect 15

[0199] In Aspect 15, the automatic document feeder according to Aspect 10 or 11 further includes an opposing member above the plate and facing the plate.

Aspect 16

[0200] In Aspect 16, in the automatic document feeder according to Aspect 13, the sound generator has a cutout in a contact face between the sound generator and the document or a rough face having a surface roughness greater than a surface roughness of the document conveyance path.

Aspect 17

[0201] In Aspect 17, the automatic document feeder according to Aspect 11 further includes a rib on the document conveyance path and extending in a conveyance direction of the document. Multiple ribs including the rib are disposed at intervals in the width direction of the document conveyance path. The sound generator has the protrusion protruding toward the surface of the document conveyance path. The sound generator is disposed to fill a gap between adjacent ribs in the width direction of the document conveyance path. A height of the one sound generator to the surface of the document conveyance path is lower than a surface of the document conveyance path.

Aspect 18

[0202] In Aspect 18, an image forming apparatus includes the automatic document feeder according to any one of Aspects 10 to 17, and an image forming device to form an image on a medium.

Aspect 19

[0203] In Aspect 19, an automatic document feeder includes a document conveyance path, a protrusion, and a sound collector. The document conveyance path is a path in which a document is conveyed to a reading position to read the document. The protrusion extends in a direction orthogonal to a conveyance direction of the document in the document conveyance path and protruding upward toward a downstream side from an upstream side in the document conveyance path of the conveyance direction of the document, to contact a surface of the document that conveys in the document conveyance path. The sound collector collects sound of the document that conveys in the document conveyance path. Conveyance of the document to be conveyed to the reading position is stopped based on the sound collected by the sound collector.

Aspect 20

[0204] In Aspect 20, in the automatic document feeder according to Aspect 19, multiple protrusions including the protrusion are disposed in the conveyance direction of the document in the document conveyance path.

Aspect 21

[0205] In Aspect 21, the automatic document feeder according to Aspect 20 further includes a rotary member to contact the document in the document conveyance path. The multiple protrusions are on both sides of the document conveyance path in a direction orthogonal to the conveyance direction of the document, interposed by the rotary member.

[0206] Aspect 22 In Aspect 22, an image forming apparatus includes the automatic document feeder according to any one of Aspects 19 to 21, and an image forming device to form an image on a medium.

[0207] The embodiments have been described above, but the embodiments are presented as examples, and the present disclosure is not limited to the embodiments. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. These embodiments and variations are included in the scope and gist of the invention and are included in the invention described in the claims and the equivalent scope thereof.

[0208] The present disclosure is not limited to specific embodiments described above, and numerous additional modifications and variations are possible in light of the teachings within the technical scope of the appended claims. It is therefore to be understood that the disclosure of this

patent specification may be practiced otherwise by those skilled in the art than as specifically described herein, and such, modifications, alternatives are within the technical scope of the appended claims. Such modifications are also included in the technical scope of the present disclosure.

[0209] The present disclosure is not limited to specific embodiments described above, and numerous additional modifications and variations are possible in light of the teachings within the technical scope of the appended claims. It is therefore to be understood that, the disclosure of this patent specification may be practiced otherwise by those skilled in the art than as specifically described herein, and such, modifications, alternatives are within the technical scope of the appended claims. Such embodiments and variations thereof are included in the scope and gist of the embodiments of the present disclosure and are included in the embodiments described in claims and the equivalent scope thereof.

[0210] The effects described in the embodiments of this disclosure are listed as the examples of preferable effects derived from this disclosure, and therefore are not intended to limit to the embodiments of this disclosure.

[0211] The embodiments described above are presented as an example to implement this disclosure. The embodiments described above are not intended to limit the scope of the invention. These novel embodiments can be implemented in various other forms, and various omissions, replacements, or changes can be made without departing from the gist of the invention. These embodiments and their variations are included in the scope and gist of this disclosure and are included in the scope of the invention recited in the claims and its equivalent.

[0212] Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

[0213] Each of the functions of the described embodiments may be implemented by one or more processing circuits or circuitry. Processing circuitry includes a programmed processor, as a processor includes circuitry. A processing circuit also includes devices such as an application specific integrated circuit (ASIC), digital signal processor (DSP), field programmable gate array (FPGA), and conventional circuit components arranged to perform the recited functions.

Claims

1. An automatic document feeder comprising: a document stacker on which a document is placed; a document conveyor to convey the document on the document stacker along a document conveyance path in a conveyance direction; a sound generator on a surface of the document conveyance path, the sound generator having a protrusion or a recess: protruding or recessed from the surface of the document conveyance path; and extending continuously or intermittently from one end to another end of the document conveyance path in a width direction orthogonal to the conveyance direction; a sound collector to collect sound generated by contact between the sound generator and a fastening object attached to the document; and circuitry configured to predict whether the fastening object is attached to the document, based on the sound collected by the sound collector.

2. The automatic document feeder according to claim 1, further comprising: multiple sound generators including the sound generator, the multiple sound generators disposed at intervals in the conveyance direction, wherein each of the multiple sound generators includes: a first face raised vertically from the surface of the document conveyance path, and a second face inclined relative to the first face and connecting an edge of the first face and the surface of the document conveyance path.

3. The automatic document feeder according to claim 2, wherein the multiple sound generators are disposed at different intervals in the conveyance direction.

4. The automatic document feeder according to claim 1, further comprising a rib on the document

conveyance path and extending in the conveyance direction, wherein the sound generator has the recess recessed from a surface of the rib.

5. The automatic document feeder according to claim 1, wherein the protrusion of the sound generator includes: a plate having a sloped face; and an elastic member movably supporting the plate in a direction orthogonal to the surface of the document conveyance path.

6. The automatic document feeder according to claim 5, further comprising an opposing member above the plate and facing the plate.

7. The automatic document feeder according to claim 1, wherein the sound generator has: a cutout in a contact face between the sound generator and the document; or a rough face having a surface roughness greater than a surface roughness of the document conveyance path.

8. The automatic document feeder according to claim 1, further comprising a rib on the document conveyance path and extending in a conveyance direction of the document, wherein multiple ribs including the rib are disposed at intervals in the width direction of the document conveyance path, the sound generator has the protrusion protruding toward the surface of the document conveyance path, the sound generator is disposed to fill a gap between adjacent ribs in the width direction of the document conveyance path, and a height of the one sound generator to the surface of the document conveyance path is lower than a surface of the document conveyance path.

9. An image forming apparatus comprising: the automatic document feeder according to claim 1; and an image forming device to form an image on a medium.

10. An automatic document feeder comprising: a document conveyance path in which a document is conveyed to a reading position to read the document; a protrusion extending in a direction orthogonal to a conveyance direction of the document in the document conveyance path and protruding upward toward a downstream side from an upstream side in the document conveyance path of the conveyance direction of the document, to contact a surface of the document that conveys in the document conveyance path; and a sound collector to collect sound of the document that conveys in the document conveyance path, wherein conveyance of the document to be conveyed to the reading position is stopped based on the sound collected by the sound collector.

11. The automatic document feeder according to claim 10, wherein multiple protrusions including the protrusion are disposed in the conveyance direction of the document in the document conveyance path.

12. The automatic document feeder according to claim 11, further comprising a rotary member to contact the document in the document conveyance path, wherein the multiple protrusions are on both sides of the document conveyance path in a direction orthogonal to the conveyance direction of the document, interposed by the rotary member.

13. An image forming apparatus comprising: the automatic document feeder according to claim 10; and an image forming device to form an image on a medium.
